Delivrd, inFlow, etc, but these operate on a freemium business model in which commercial versions have a user fee, whereas Apache OFBiz is freely available for download. Thus, Apache OFBiz (Figure 3) is an ideal choice for inventory management.

Remote sensing software

Remote sensing is the process by which information about an object or place is obtained without making any physical contact, and is largely used in areas like meteorology, oceanography, geology, glaciology, etc. If accurate information is available in these fields, it can be used to predict and thereby reduce the impact of many natural calamities. Remote sensing can also be used in a post-disaster environment. For example, it can be used to find alternative routes to reach areas isolated due to floods, volcanic eruptions, etc.

Figure 4: Logo of Opticks

Remote sensing involves passive sensing and active sensing. The former detects radiation emitted from the observed object or place. Even though passive remote sensing often involves photography, a lot of information can also be obtained by using techniques like infra-red radiation detection, radio wave detection, etc. In active sensing, an energy source is used to scan the object or place to be sensed. Analysing the massive data sets produced by passive and active sensing is a tedious and error-prone task. Due to this, a lot of geospatial toolkits are used to analyse remotely sensed data. Remote sensing software often acts like image processing software.

Some of the popular remote sensing software have proprietary licences. But we will only discuss open source remote sensing software. I really wanted to include Google Earth in this list. But remembering the animosity of many in the open source community towards freeware, I will include Google Earth Enterprise, a geospatial application licensed under the Apache 2.0 licence. It provides the ability to build and host custom made two-dimensional maps and 3D globes. Another popular tool is OpenEV, a geospatial toolkit capable of processing remote sensing data to provide 3D elevation data and geo-referencing, which is the process by which a point in a geo-sensed image is mapped to an actual location. Opticks (Figure 4) is another popular remote sensing application that can process both remote sensed images and videos. It also has the advantage of additional plugins for further extensions.

Geographic information systems

A geographic information system (GIS) is a tool used to store and analyse spatial and geographic data. A GIS can be used to highlight a cluster of points in a geographic area. One of the earliest recorded uses of a GIS was during the cholera epidemic of Paris in 1832, when the locations of people affected with the disease were marked on a map to identify the source of the epidemic. In a post-disaster environment, there is always a high probability of diseases spreading, particularly when water sources are contaminated, like during a flood. In such situations, a GIS has a huge role to play. Now let us discuss a few popular open source GIS.

QGIS is a cross-platform, open source GIS that allows the comparison and export of hard copy maps in a digital form. It also allows analysis and editing of spatial information. PostGIS is a spatial database extender for PostgreSQL, an object-relational database management system. It allows location queries to be run in SQL over geographic objects. PostGIS is released under the GNU General Public License. Another popular yet proprietary GIS is ArcGIS (Figure 5), which is used for creating and using maps, as well as for analysing such mapped information. ArcGIS also helps in managing geographic information in a database.

Figure 5: Logo of ArcGIS

I would like to point out one fact before concluding this article. All the ideas mentioned here are from a layman's perspective. Rather than vouching for the 100 per cent efficacy of any particular idea or solution, I instead hope this article will lead to a debate about speedy solutions for disaster management, which will eventually lead to better ideas than the ones presented here. However, I strongly believe that the proper use of computing technology can reduce the severity of natural disasters anywhere in the world. During such periods, governments and relief organisations are hard pressed for funds, so adopting open source technologies is the logical choice. Ideally, disaster management agencies should have a volunteer task force equipped with the necessary computing skills ready, so that it can be used to manage a post-disaster environment efficiently. END

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